

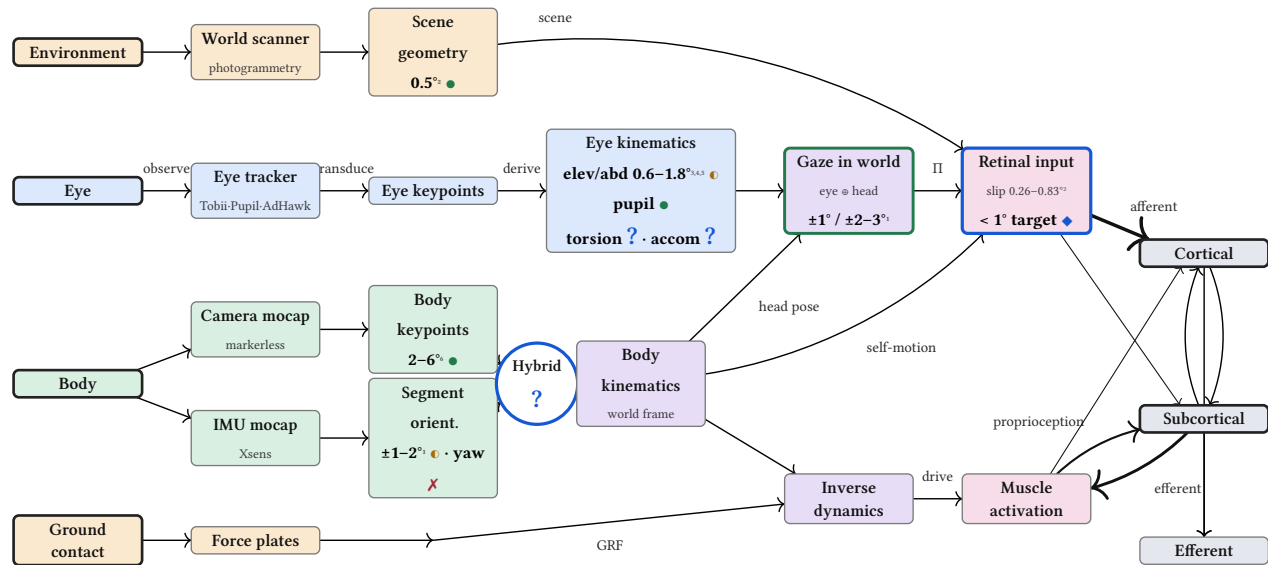


Figure 1: **Quantifying the technology landscape.** Each reported error is placed on the **quantity it measures**, not the instrument, and tagged by provenance — ● open & traceable, ● closed vendor model (uncertainty unknown), ✗ closed & **not truth-preserving** (Xsens¹ fills an out-of-bounds magnetometer yaw with a model fit), ◆ DOME target, and a big blue ? for what is **not built or measured yet** — the capabilities DOME adds. Mobile eye trackers resolve gaze (elevation/abduction) in the **head frame** to 0.6° (Tobii Pro Glasses 3⁴), 1.3–1.8° (Pupil Labs Neon³), and ≈1° (AdHawk MindLink⁵), and give pupil diameter — but **torsion and accommodation are measured by no mobile tracker** (?), and the camera–IMU **Hybrid** (?) is itself unbuilt. The quantity vision needs is gaze **in the world**: composing eye with head/body pose (IMU segment orientation ±1–2°¹ markerless keypoints 2–6°⁶) and photogrammetric scene geometry (median ≈3 cm ≈ 0.5°²), the only published end-to-end eye + IMU + world method — **this team’s own prior work** — reaches ±1° calibrated and ±2–3° during natural walking (Matthis, Yates & Hayhoe 2018¹), with retinal-slip precision mode 0.26° / median 0.83° (Muller et al. 2022/2023²), against the < 1° gaze-in-world needed to predict neural activity. No existing toolchain delivers sub-1° gaze-in-world with a propagated uncertainty budget — the DOME goal.

Sources: ¹ Matthis, Yates & Hayhoe 2018, *Curr. Biol.* 28:1224–1233 (this team); ¹ Xsens/Movella MVN magnetometer-mapping behavior; ² Muller et al. 2022 (*bioRxiv*) / 2023, *eLife* 12:e82410 (this team); ³ Pupil Labs Neon accuracy report; ⁴ Tobii Pro Glasses 3 spec; ⁵ AdHawk MindLink spec; ⁶ Cherian 2026 FreeMoCap dissertation, cf. Theia3D & OpenCap validations.